Paper Review:
Temporal Convolutional Network for Type-1 Diabetes

Zaidi, S. M. A., Chandola, V., Ibrahim, M., Romanski, B., Mastrandrea, L. D., and Singh, T. (2021). Multi-step ahead predictive model for blood glucose concentrations of type-1 diabetic patients. Scientific reports, 11(1), 24332.

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PCELab
Predictive Computational Engineering

Motivation

1. Patients with type 1 diabetes (T1D) must monitor their BG level regularly to take necessary actions to prevent hypoglycemia (low BG levels) and hyperglycemia (high BG levels).
2. Factors that affect future BG levels: prior BG levels, insulin (basal and bolus) dose, meal intake, body mass index, physical activity, etc.
3. The drawbacks of Recurrent neural networks (RNN) like Long Short-Term Memory (LSTM) for sequence modeling are
   - Difficulty in memorizing very long temporal sequences due to poor propagation of information between cells.
   - Vanishing and exploding gradients.
   - Time-consuming training process.
4. advantages of Temporal Convolutional Networks (TCN) for modeling temporal sequences.
   - Coupling 1D convolution operations with dilated filters captures a larger receptive field of the input sequence.
   - Better than RNN at modeling long-range temporal dependencies.

Modeling Problem

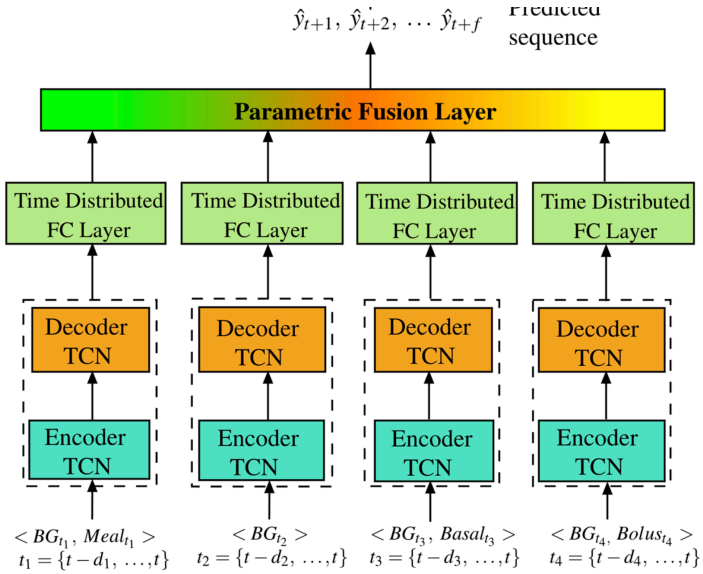
T1D patient data: The datasets are adopted from Tidepool (https://www.tidepool.org/) and consist of 97 patients with blood glucose recorded every 5 mins.

Problem statement. Given the history data of *meal intake* (M), *blood glucose* (BG), *basal insulin rate* (Ba), and *bolus insulin* (Bo), predict the future BG values in a specific patient:

- Given $\{\mathrm{M}_0, \ldots, \mathrm{M}_t\}; \{\mathrm{BG}_0, \ldots, \mathrm{BG}_t\}; \{\mathrm{Ba}_0, \ldots, \mathrm{Ba}_t\}$ and $\{\mathrm{Bo}_0, \ldots, \mathrm{Bo}_t\}$
- Predict $\mathrm{BG}_{t+1}, \ldots, \mathrm{BG}_{t+f}$, where f is the prediction horizon of 30 mins.

Model formulation. This study accounts for the different input history lengths.

- Given $\{(\mathrm{M}, \mathrm{BG})_{t-p_1}, \ldots, (\mathrm{M}, \mathrm{BG})_t\}$ with $p_1 = 3h$
  $\{\mathrm{BG}_{t-p_2}, \ldots, \mathrm{BG}_t\}$ with $p_2 = 30min$
  $\{(\mathrm{Ba}, \mathrm{BG})_{t-p_3}, \ldots, (\mathrm{Ba}, \mathrm{BG})_t\}$ with $p_3 = 3h$
  $\{(\mathrm{Bo}, \mathrm{BG})_{t-p_4}, \ldots, (\mathrm{Bo}, \mathrm{BG})_t\}$ with $p_4 = 3h$
- Predict $\mathrm{BG}_{t+1}, \ldots, \mathrm{BG}_{t+f}$,



Temporal convolutional network (TCN)

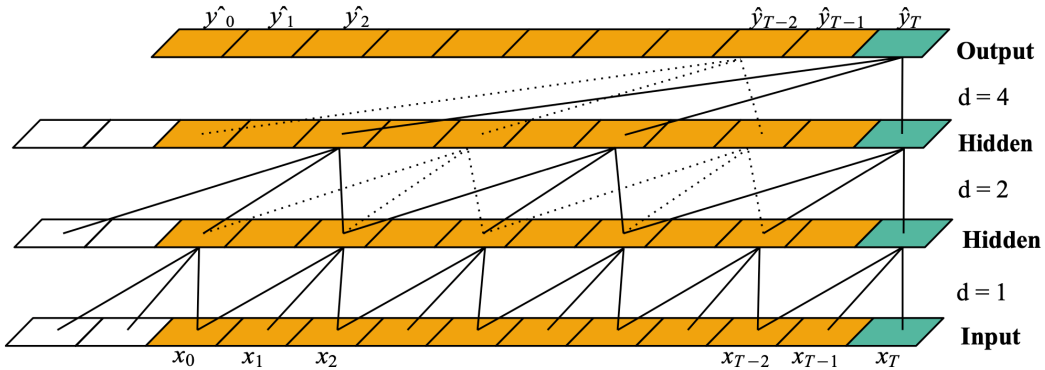


Figure: Dilated causal convolution, k=3

Causal convolution:

- The convolution operation output at time t in the current layer considers elements at time t and earlier in the previous layer.
- Ensures no information leakage from the future to the present when computing the output at t.
- Limitation: looking back at the history of the sequence with a linear size in the network's depth makes it challenging for longer input sequences. Dilated convolution is used to alleviate this challenge.

Temporal convolutional network (TCN)

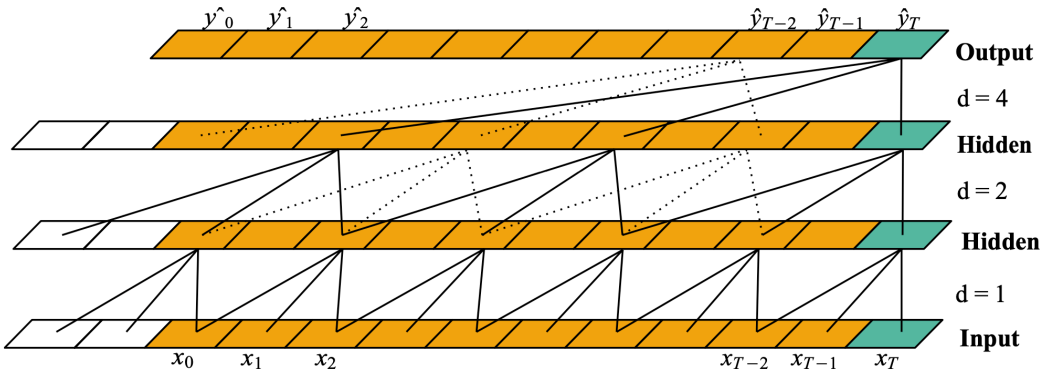


Figure: Dilated causal convolution, k=3

Dilated convolution:

- Enables a large receptive field, allowing the model to consider a broader range of input sequences efficiently (without increasing the model size by using a larger filter size).
- Applies the convolution filter to a larger area than its size by expanding the filter with zeros (dilation), thus skipping certain input elements according to a dilation factor.
- Each convolution output carries rich information while increasing the dilation factor extends the range of inputs considered, expanding the receptive field.
- Dilated convolution operation: $F(s) = (\mathbf{x} *_d f) = \sum_{i=0}^{k-1} f(i) \cdot \mathbf{x}_{s-d.i}$, where $\mathbf{x}$: input sequence, d : dilation factor, f : filter, k : filter size, s : an element of the input sequence, $\mathbf{x}$, $s - d.i$ past direction.

Temporal convolutional network (TCN)

Residual network.

- Using deeper and larger TCNs helps compute larger receptive fields.
- To ensure stability throughout training, avoiding issues like vanishing or exploding gradients, residual blocks are incorporated into the TCN-based model[1].
- In residual blocks, layers learn the residual function $F(x) = H(x) - x$, and the output of the residual block will lead to the learning of the desired mapping function $H(x)$.
- This is enabled by shortcut/skip connections, which skip one or more layers without adding extra parameters, thus avoiding increased computational complexity.

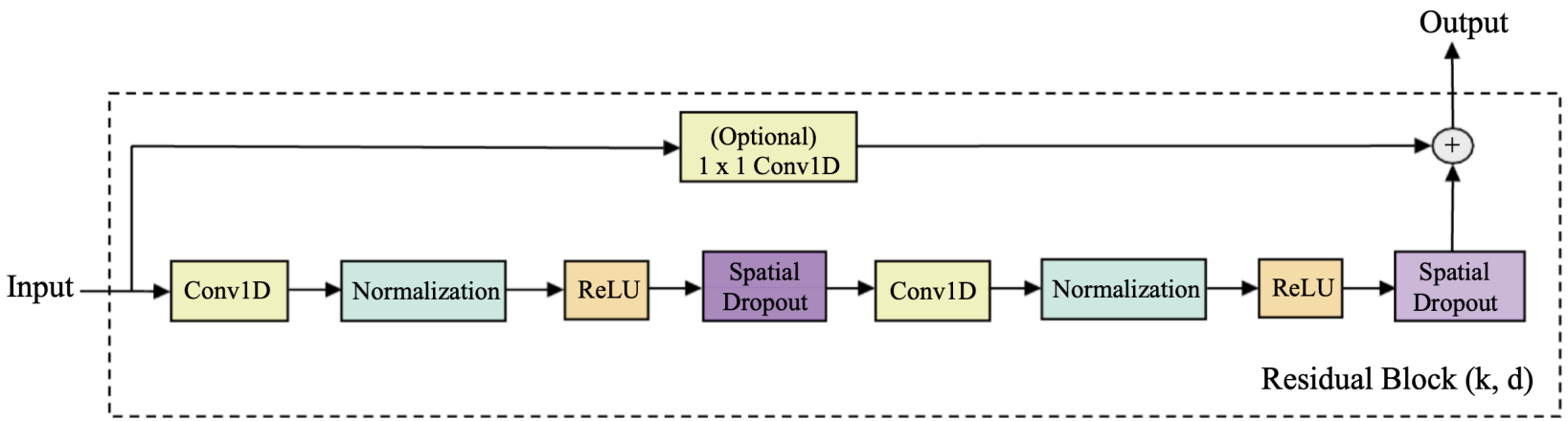


Figure: Residual block in TCN. Each "Conv1D" is a dilated casual convolution shown in the previous slide. The 1×1 conv1D layer ensures that the width of the output matches the input.

[1] Bai, S., Kolter, J. Z., and Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271.

Sequence-to-Sequence model

- **TCN Encoder:** Processes input sequence, capturing short-term and long-term dependencies with dilated causal convolutions. Outputs a latent state vector that encapsulates the temporal patterns and dependencies of the input sequence.
- **TCN Decoder:** Uses latent states to generate future time series values step-by-step. Each prediction becomes an input for the next, using convolutions to model dependencies and ensuring predictions are influenced by past values.
- **Advantages:** Captures long-term dependencies effectively with dilated convolutions without needing many layers. Causal convolutions ensure predictions rely only on past values, crucial for accurate time series forecasting.

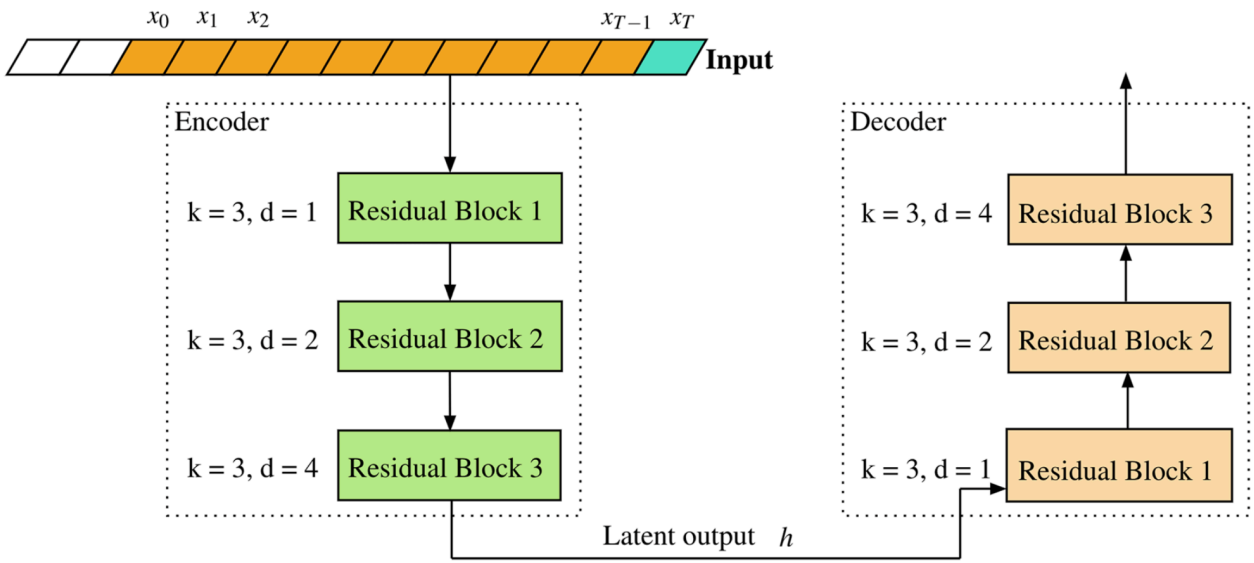


Figure: Sequence to sequence modeling architecture with encoder and decoder

Full Model

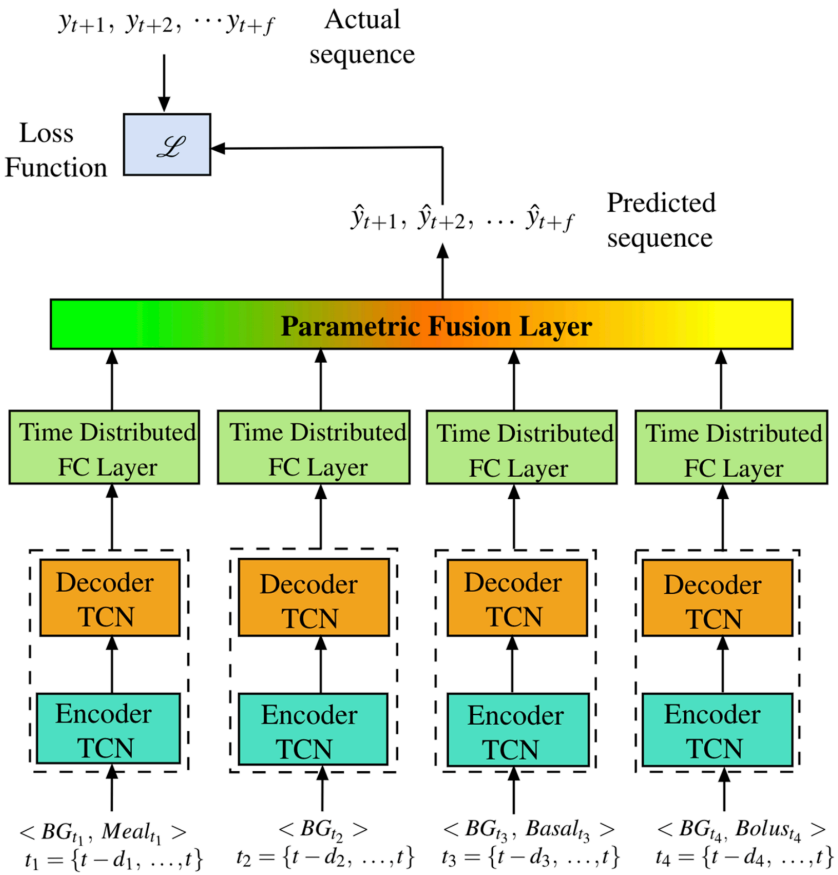


Figure: Proposed full Model with four different TCNs

The fully connected layer integrates the decoder's output sequence into the parametric that combines four sequences from different inputs, producing the final output such that,

$$\hat{\mathbf{Y}} = \mathbf{W}_1 \odot \hat{\mathbf{Y}}_1 + \mathbf{W}_2 \odot \hat{\mathbf{Y}}_2 + \mathbf{W}_3 \odot \hat{\mathbf{Y}}_3 + \mathbf{W}_4 \odot \hat{\mathbf{Y}}_4,$$

where $\hat{\mathbf{Y}}_i$ is the output sequence and $\mathbf{W}_i$ is the trainable weight parameters.

Full Model

Loss function: Huber loss with tunable hyperparameter $\delta = 0.5$

$$L_\delta(\mathbf{y}, \hat{\mathbf{y}}) = \begin{cases} \frac{1}{2}(\mathbf{y} - \hat{\mathbf{y}})^2, & \text{for } |\mathbf{y} - \hat{\mathbf{y}}| \leq \delta \\ \delta\left(|\mathbf{y} - \hat{\mathbf{y}}| - \frac{1}{2}\delta\right), & \text{otherwise} \end{cases}$$

Weighted penalties are applied to improve predictions and capture rare events like hypoglycemia and hyperglycemia. The cost of incorrect hypoglycemic values is heavily penalized by multiplying it with weight $\lambda_{hypo} = 100$. Similarly, the weight of hyperglycemic is $\lambda_{hypo} = 10$, and normoglycemic values are minimally penalized by $\lambda_{hypo} = 1$.

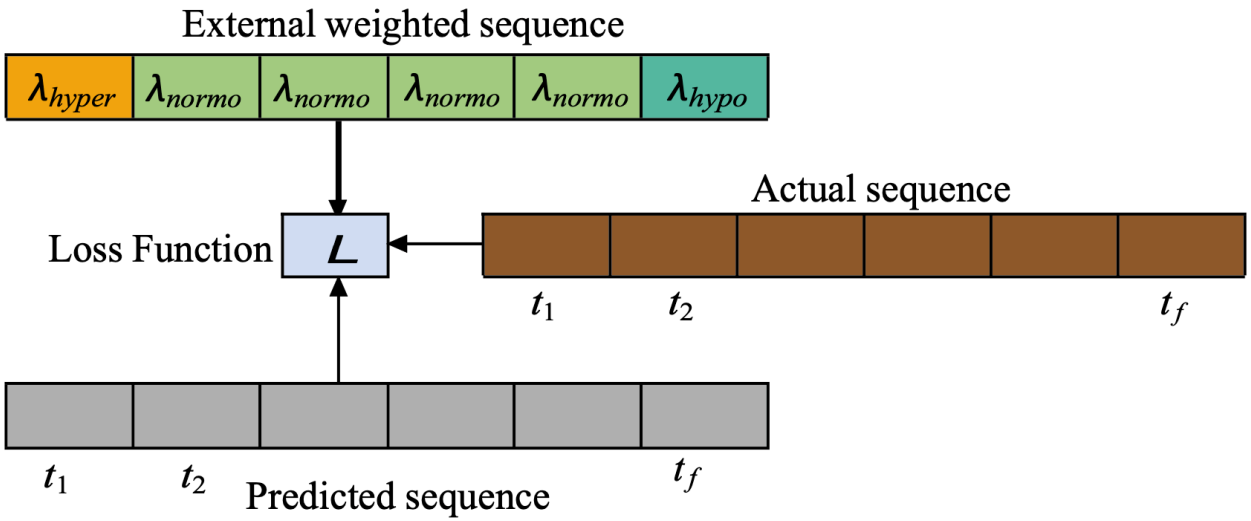
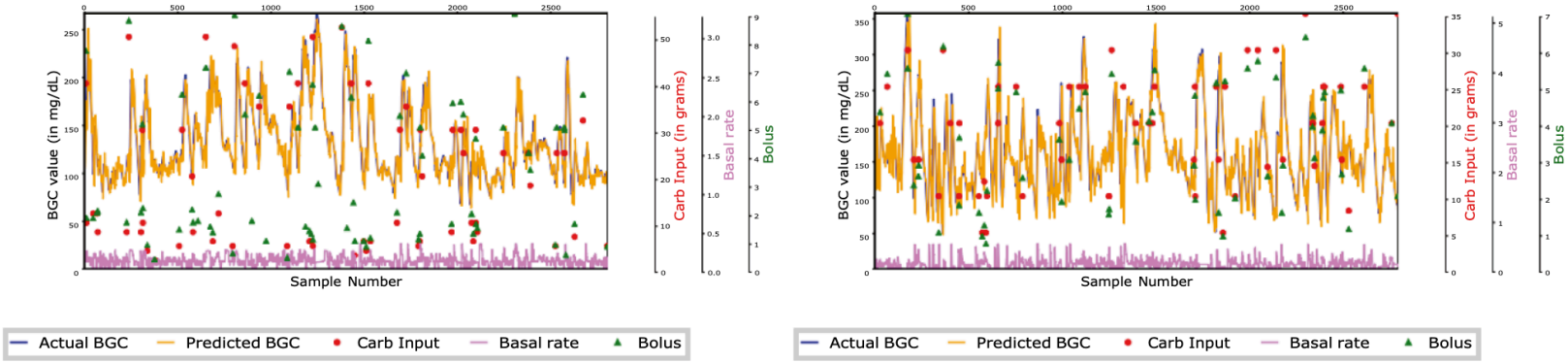


Figure: Loss function update with external weights for incorrect hypoglycemic, hypoglycemic and normoglycemic events

Results



(a) Patient 77 **(b)** Patient 96

Figure: 30 mins forecasting on validation data for two T1D patients.

Sensitivity of features and past history (p) on BG prediction.

- 1st component (M, BG): Turning off meal intake M slightly decreases prediction performance. RMSE decreases when both meal intake and past history ($p_1 = 0$) are switched off. No clear trend with increasing history.
- 2nd component (BG): Slight decrease in model performance as BG switched off
- 3rd component (Ba, BG): Slight decrease in RMSE with both Ba and BG are off
- 4th component (Bo, BG): RMSE decreases when both BG and bolus insulin are switched off. Slight decrease in performance when only bolus insulin is switched off.
- The contribution of the 1st (3 h history of (M, BG)) and the 2nd (30 mins of (BG)) to be more on the predictions in comparison to the 3rd and 4th components (3 h of (Ba, BG)) and the fourth component (3 h of (Bo, BG)).

Extension to Bayesian training

Bayesian Inference

$$\pi_{post}(\boldsymbol{\theta}|\boldsymbol{D}) \propto \pi_{post}(\boldsymbol{D}|\boldsymbol{\theta})\pi_{prior}(\boldsymbol{\theta})$$

- Parameters $\boldsymbol{\theta}$ weights within the encoder and decoder convolutional filters of TCN layers, time-distributed fully-connected layers, and the learnable parameters for fusing the output sequences.
- Solution: Variational Inference via Bayes by Backpropagation.
- Critical to the Bayesian TCN solution is maintaining the structure of dilated convolutions and preserving temporal dependencies[2].

Network sparsification: eliminating irrelevant weights for model reduction.

- apply a sparsity-enforcing prior via the Hard Concrete distribution which approximates L0 norm regularization in a tractable manner[3], followed by magnitude-based thresholding.
- Solution: Variational Inference via Bayes by Backpropagation.
- Critical to the Bayesian TCN is maintaining the Causal structure and temporal coherence. This can be achieved by, for example, appropriate zero-padding of input sequences, ensuring convolution operations remain causal without altering receptive fields or causal connections.

[2] Hong, S., He, Y., Zhang, J., Jiang, C., Deng, Y. (2022, April). Remaining Useful Life Prediction via Bayesian Temporal Convolutional Networks. In 2021 International Conference on Advanced Computing and Endogenous Security. IEEE.

[3] Louizos, C., Welling, M., Kingma, D. P. (2017). Learning sparse neural networks through L_0 regularization. arXiv preprint arXiv:1712.01312.